

Tao Wang

Doctoral Candidate

Nanjing University of Aeronautics and Astronautics, China

wangtao21@nuaa.edu.cn • taow1776@gmail.com • daizigege.github.io • [Google Scholar Link](#)

Education

- **Ph.D.**, Cyberspace Security, Nanjing University of Aeronautics and Astronautics 2024.04–2027.04
- **Visiting Ph.D.**, Cyberspace Security, University of Milan, SPDP Lab 2025.09–2026.09
- **M.S.**, Cyberspace Security, Nanjing University of Aeronautics and Astronautics 2021.09–2024.04
- **B.E.**, Computer Science and Technology, Anhui Normal University 2017.09–2021.06

Research Interests

- **Field:** Data privacy, computer vision, and artificial intelligence
- **Focus:** Implicit privacy, usable privacy, visual privacy, and face privacy

Previous Research: Controllable Identity Visibility in Usable Face Privacy:

- **Universal Privacy Protection:** Developed an information-theoretic face anonymization framework that removes identity information from both human and machine observers [C1].
- **Socially-Aware Privacy:** Proposed a de-identification approach that preserves recognizability for familiar individuals while preventing machine recognition [J5].
- **Human–AI Privacy Asymmetry:** Designed an identity hider that leverages differences between human and machine perception, only enabling machine utility [J6].
- **Authorized Identity Access:** Developed a cancelable face generation framework that supports identity recognition exclusively by authorized systems [C2].

Current and Future Research: Human-Centered Implicit Privacy Governance in the GenAI Era. As GenAI systems increasingly infer sensitive attributes from seemingly innocuous information, privacy risks are becoming implicit, relational, and difficult for individuals to perceive or control. We aim to develop graph-based frameworks for implicit privacy discovery, risk awareness, privacy-preserving interventions, and accountability, while investigating how such risks shape user behavior, social norms, and regulatory responses.

Publications

- **Statistics:** Google scholar citations > 423, h-index = 9; published 7 papers as first or corresponding author, including 3 in CCF-A, 1 with impact factor > 10, and 1 highly cited in ESI
- **Representative publications:** [J7], [J2], and [J1] in *IEEE Transactions on Dependable and Secure Computing*; [J4] and [J10] in *IEEE Transactions on Information Forensics and Security* ; [J8] in *ACM Computing Surveys*; [J3] in *IEEE Transactions on Pattern Analysis and Machine Intelligence*; [C2, C3] in *ACM International Conference on Multimedia*
- †: Corresponding author

Refereed Journal Papers

- J1. Wen, W., Wu, Z., Zhang, Y., **Wang, Tao**, Xiao, X. & Fang, Y. Medical Archive in an Image: Generating a High-Capacity Customizable Cover Image for Medical Privacy Protection. *IEEE Transactions on Dependable and Secure Computing* **23**, 4735–4747 (2026).
- J2. Xu, K., Wen, W., Qi, S., **Wang, Tao**, Zhang, Y. & Fang, Y. Detecting Malicious Concepts without Image Generation in AI-Generated Content (AIGC). *IEEE Transactions on Dependable and Secure Computing*, just accepted (2026).
- J3. Zhang, Y., Ji, J., **Wang, Tao**, Zhao, R., Wen, W. & Xiang, Y. Make Identity Indistinguishable: Utility-Preserving Face Dataset Publication With Provable Privacy Guarantees. *IEEE Transactions on Pattern Analysis and Machine Intelligence* **48**, 127–139 (2026).
- J4. **Wang, Tao**, Wen, W., Xiao, X., Hua, Z., Zhang, Y. & Fang, Y. Beyond Privacy: Generating Privacy-Preserving Faces Supporting Robust Image Authentication. *IEEE Transactions on Information Forensics and Security* **20**, 2564–2576 (2025).
- J5. **Wang, Tao**, Zhang, Y., Xiao, X., Xu, K., Yuan, L., Wen, W. & Fang, Y. Make Identity Unextractable yet Perceptible: Synthesis-Based Privacy Protection for Subject Faces in Photos. *arXiv preprint arXiv:2509.11249* (2025).
- J6. **Wang, Tao**, Zhang, Y., Yang, Z., Xiao, X., Zhang, H. & Hua, Z. Seeing is Not Believing: An Identity Hider for Human Vision Privacy Protection. *IEEE Transactions on Biometrics, Behavior, and Identity Science* **7**, 170–181 (2025).
- J7. Zhang, Y., Yang, Z., **Wang, Tao**[†], Hua, Z. & Weng, J. Tailor-Made Face Privacy Protection via Class-Wise Targeted Universal Adversarial Perturbations. *IEEE Transactions on Dependable and Secure Computing* **22**, 5108–5120 (2025).
- J8. **Wang, Tao**, Zhang, Y., Qi, S., Zhao, R., Xia, Z. & Weng, J. Security and Privacy on Generative Data in AIGC: A Survey. *ACM Computing Surveys* **57**, 1–34 (2024).
- J9. Xie, Y., Zhou, Y., **Wang, Tao**[†], Wen, W., Yi, S. & Zhang, Y. Reversible Gender Privacy Enhancement via Adversarial Perturbations. *Neural Networks* **172**, 106130 (2024).
- J10. Zhang, Y., **Wang, Tao**, Zhao, R., Wen, W. & Zhu, Y. RAPP: Reversible Privacy Preservation for Various Face Attributes. *IEEE Transactions on Information Forensics and Security* **18**, 3074–3087 (2023).

Refereed Conference Papers

- C1. **Wang, Tao**. *Information-Theoretic Face Anonymization* in Under Review (2026).
- C2. **Wang, Tao**, Zhang, Y., Xiao, X., Yuan, L., Xia, Z. & Weng, J. *Make Privacy Renewable! Generating Privacy-Preserving Faces Supporting Cancelable Biometric Recognition* in *Proceedings of the ACM International Conference on Multimedia* (2024), 10268–10276.
- C3. Yang, Z., Zhang, Y., **Wang, Tao**, Hua, Z., Xia, Z. & Weng, J. *Once-for-all: Efficient Visual Face Privacy Protection via Person-Specific Veils* in *Proceedings of the ACM International Conference on Multimedia* (2024), 7705–7713.

Awards & Honors

- Doctoral Student Program of the Young Sci-Tech Talents Cultivation Project, CAST, 2025
- Puxin Elite Scholarship, 2025
- CSC Scholarship of High-Level Postgraduate Program, 2025
- ACM MM Honorable Mention Award, 2024
- National Scholarships for Postgraduate Students, 2023

Research Grants

- **PI**, Postgraduate Research Innovation Program, Jiangsu Provincial Department of Education; Friendly Face Privacy Protection CN¥ 15,000 2025–2027

Presentations & Talks

2026 **MILAN0DAY**, University of Milan. Invited Talk: “AI and the Death of Privacy? Navigating New Threats and Modern Defenses.”

Professional Services

Journal Reviewers

TKDE, TDSC, TCSVT, TNSE, TBIOM, TOMM, EAAI, ESWA, KBS, IOTJ.

Conference Reviewer

ICLR, SIGKDD, ACM MM

Membership

ACM EIGTRUST Membership; CSIG Membership

References

- **Pierangela Samarati**, General supervisor, pierangela.samarati@unimi.it , University of Milan
- **Yushu Zhang**, Ph.D. supervisor, zhangyushu@jxufe.edu.cn, Jiangxi University of Finance and Economics
- **Shuren Qi**, General supervisor, shurenqi@nuaa.edu.cn, City University of Hong Kong
- **Liping Sun**, B.E. supervisor, slp620@163.com, Anhui Normal University

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